

Karthik Raja Nichenametla

+91 9494999834 | nkarthikraja46@gmail.com | LinkedIn | GitHub | Portfolio

SUMMARY

SDET and automation engineer with hands-on experience across CI/CD pipeline integration, containerized test environments, and AI/LLM-assisted automation workflows. Proficient in Python scripting, GitHub Actions CI/CD, Docker, and Kubernetes fundamentals, with applied experience building an LLM-based automation pipeline (LangChain) and structured REST API validation. Skilled at triaging pipeline and test failures through root cause analysis to improve delivery reliability.

EXPERIENCE

Software Development Engineer in Test (SDET)

IMTF

Jun 2026 – Present

Bengaluru, Karnataka

- Develop and maintain UI and API test automation frameworks using Playwright, TypeScript, and JavaScript, integrated as components within CI/CD pipelines.
- Automate REST API testing and maintain CI/CD pipeline integrations in GitHub Actions to support regression and release validation.
- Operate and maintain Docker- and Kubernetes-based test environments, applying containerization fundamentals to support pipeline execution.
- Monitor test-pipeline health and triage recurring failures through root cause analysis, improving CI/CD reliability and reducing false-positive escalations.
- Use Python, SQL, Linux, and Git/GitHub for day-to-day automation scripting, environment troubleshooting, and version control.

SDET / QA Automation Intern

Bluecopa

Sep 2025 – Mar 2026

Bengaluru, Karnataka

- Designed, developed, and maintained automated test suites using Playwright (TypeScript) and Testsigma, covering end-to-end, regression, and cross-browser testing for critical user journeys.
- Built and executed REST API test automation for 15+ endpoints using Pytest, Postman, and JSON Schema validation, covering functional, negative, and edge-case scenarios.
- Implemented and maintained CI/CD pipeline integrations in GitHub Actions, reducing release regression issues by 25% and enabling parallel test execution.
- Debugged and stabilized flaky tests through root cause analysis, reducing false-positive failures by 40%.
- Collaborated with developers in Agile/Scrum teams for bug identification, reporting, validation, and early quality assurance.

Backend & Data Science Intern

Toolika Innovations LLP

Feb 2025 – Aug 2025

Gurugram, Haryana

- Developed and validated 3+ production-grade REST APIs using FastAPI, SQLAlchemy, and Pydantic, improving data retrieval performance by 20%.
- Engineered an LLM-based document extraction pipeline (LangChain, Python) with hierarchical chunking and few-shot prompting, increasing data ingestion efficiency by 25%.
- Implemented structured schema validation and prompt templates to ensure consistent outputs for downstream analytics workflows, reducing manual data handling across teams.

TECHNICAL SKILLS

Programming: Python, TypeScript, JavaScript, SQL

CI/CD & Pipeline Automation: GitHub Actions, CI/CD Pipeline Integration & Maintenance, Automated Test Execution, Build Validation, Pipeline Failure Triage

Platforms & Infrastructure: Docker, Kubernetes (pods, deployments, services), Linux, Git/GitHub

AI/LLM-Assisted Automation: LangChain, Prompt Engineering, Few-Shot Prompting, Structured Output Validation

Test Automation: Playwright (TypeScript), End-to-End Testing, UI Automation, API Testing, Regression Testing, Cross-Browser Testing, Schema Validation, Pytest, Testsigma, Allure Reporting

API & Backend: REST APIs, Postman, FastAPI, Pydantic, SQLAlchemy, Request/Response Validation

Quality Engineering: Debugging, Root Cause Analysis, Bug Reporting & Validation, STLC, SDLC, Agile/Scrum Methodologies

EDUCATION

BML Munjal University

B.Tech in Computer Science

- CGPA: 7.77/10

Sep 2021 – Jul 2025

Gurugram, Haryana